

1. Given an encoded string, return its decoded string.

The encoding rule is $k[\text{encoded_string}]$, where the string inside the square brackets is repeated exactly k times. The value of k is always a positive integer. Nested encodings may occur.

Input Format

A single line containing the encoded string.

Output Format

Print the decoded string.

Sample Input 1

3[a]2[bc]

Sample Output 1

aaabcbc

Sample Input 2

3[a2[c]]

Sample Output 2

accaccacc

For example:

Input	Result
3[a]2[bc]	aaabcbc
3[a2[c]]	accaccacc

2. Problem Statement

A square matrix is called a Z-Matrix when:

- Every element in the first row is non-zero.
- Every element in the last row is non-zero.
- Every element on the right-to-left diagonal is non-zero.
- Every other element is zero.

Input Format

n
matrix elements in n rows

Output Format

Print **True** if the matrix is a Z-Matrix; otherwise print **False**.

For example:

Input	Result
3 1 2 3 0 1 0 1 2 2	true
2 1 1 0 0	false

3. Problem Statement

Given a two-dimensional grid of uppercase characters and a target string, find a path that forms the target string.

You may move only to an adjacent cell in the following fixed order: **left, down, right, up**. A cell cannot be reused in the same path.

Starting cells are checked from the top row to the bottom row and from left to right. Print the first valid path found. Coordinates must use **1-based indexing**.

Input Format

rows columns

row1

row2

...

rowN

target

Each grid row is supplied as a continuous string without spaces.

Output Format

Print the coordinates in the format `[[r1, c1], [r2, c2], ...]`. If no path exists, print **Path not found**.

Sample Input 1

4 5

AZTEC

OZONE

HOMES

SOLID

ZOHO

Sample Output 1

[[2, 2], [2, 1], [3, 1], [3, 2]]

Sample Input 2

3 3

ABC

DEF

GHI

ABE

Sample Output 2

[[1, 1], [1, 2], [2, 2]]

For example:

Input	Result
4 5 A Z T E C O Z O N E H O M E S S O L I D Z O H O	[[2, 2], [3, 2], [3, 1], [2, 1]]
3 3 A B C D E F G H I A B E	[[1, 1], [1, 2], [2, 2]]